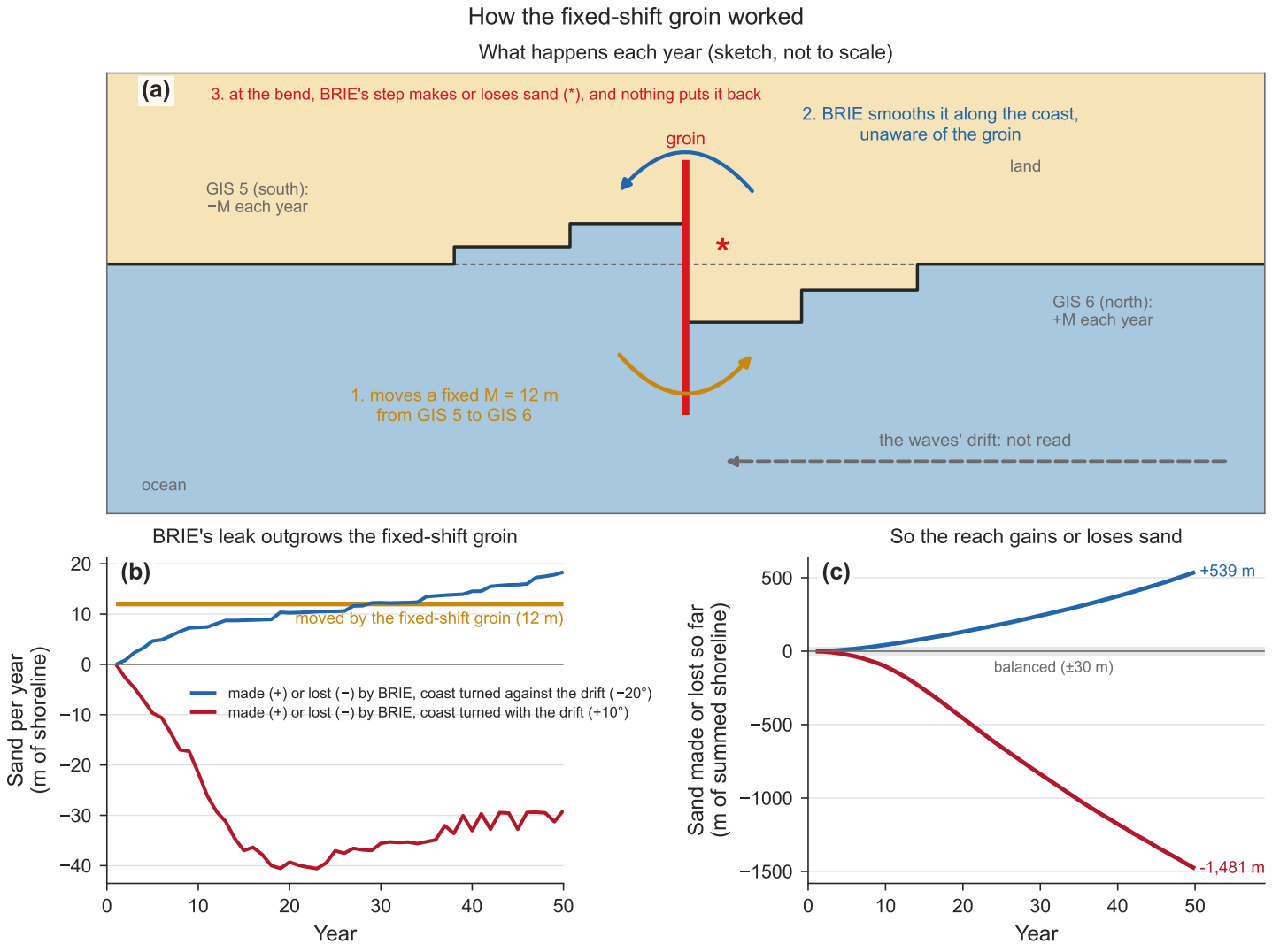
How the fixed-shift groin worked
What happens each year (sketch, not to scale)
(a)
3. at the bend, BRIE's step makes or loses sand (*), and nothing puts it back
2. BRIE smooths it along the coast, unaware of the groin
groin
land
GIS 5 (south): −M each year
*
GIS 6 (north): +M each year
1. moves a fixed M = 12 m from GIS 5 to GIS 6
the waves' drift: not read
ocean
BRIE's leak outgrows the fixed-shift groin
(b)
moved by the fixed-shift groin (12 m)
made (+) or lost (−) by BRIE, coast turned against the drift (−20°)
made (+) or lost (−) by BRIE, coast turned with the drift (+10°)
Sand per year (m of shoreline)
Year
So the reach gains or loses sand
(c)
+539 m
balanced (±30 m)
-1,481 m
Sand made or lost so far (m of summed shoreline)
Year